

Calculus I Trigonometric Identities to Memorize

Larson 12th / Cengage-focused study sheet

1. Reciprocal Identities

$$\csc x = \frac{1}{\sin x} \quad \sin x = \frac{1}{\csc x}$$

$$\sec x = \frac{1}{\cos x} \quad \cos x = \frac{1}{\sec x}$$

$$\cot x = \frac{1}{\tan x} \quad \tan x = \frac{1}{\cot x}$$

2. Quotient Identities

$$\tan x = \frac{\sin x}{\cos x} \quad \cot x = \frac{\cos x}{\sin x}$$

3. Pythagorean Identities

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

4. Useful Rearrangements

$$1 - \sin^2 x = \cos^2 x \quad 1 - \cos^2 x = \sin^2 x$$

$$\sec^2 x - 1 = \tan^2 x \quad \csc^2 x - 1 = \cot^2 x$$

$$1 - \sec^2 x = -\tan^2 x \quad 1 - \csc^2 x = -\cot^2 x$$

5. Even / Odd Identities

$$\sin(-x) = -\sin x \quad \cos(-x) = \cos x$$

$$\tan(-x) = -\tan x \quad \csc(-x) = -\csc x$$

$$\sec(-x) = \sec x \quad \cot(-x) = -\cot x$$

Memory rule: cos and sec are even; the other four trig functions are odd.

6. Double-Angle Identities

$$\sin(2x) = 2 \sin x \cos x$$

$$\cos(2x) = \cos^2 x - \sin^2 x$$

$$\cos(2x) = 2 \cos^2 x - 1$$

$$\cos(2x) = 1 - 2 \sin^2 x$$

$$\tan(2x) = \frac{2 \tan x}{1 - \tan^2 x}$$

7. Power-Reduction / Half-Angle Forms

$$\sin^2 x = \frac{1 - \cos(2x)}{2}$$

$$\cos^2 x = \frac{1 + \cos(2x)}{2}$$

8. Sum and Difference Identities

$$\sin(a + b) = \sin a \cos b + \cos a \sin b$$

$$\sin(a - b) = \sin a \cos b - \cos a \sin b$$

$$\cos(a + b) = \cos a \cos b - \sin a \sin b$$

$$\cos(a - b) = \cos a \cos b + \sin a \sin b$$

$$\tan(a + b) = \frac{\tan a + \tan b}{1 - \tan a \tan b}$$

$$\tan(a - b) = \frac{\tan a - \tan b}{1 + \tan a \tan b}$$

9. Cofunction Identities

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x$$

$$\cos\left(\frac{\pi}{2} - x\right) = \sin x$$

$$\tan\left(\frac{\pi}{2} - x\right) = \cot x$$

$$\sec\left(\frac{\pi}{2} - x\right) = \csc x$$

10. Core Seven - Highest Priority Review

$$\tan x = \frac{\sin x}{\cos x} \quad \cot x = \frac{\cos x}{\sin x}$$

$$\sec x = \frac{1}{\cos x} \quad \csc x = \frac{1}{\sin x}$$

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

Also Memorize: Special Angles

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin x$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan x$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undef.

Unit-Circle Quadrant Signs

QI	all trig functions positive
QII	sine / cosecant positive
QIII	tangent / cotangent positive
QIV	cosine / secant positive

Fast Recognition for Calculus

- Convert tan, cot, sec, csc to sine/cosine when simplification stalls.
- Look first for $\sin^2 x + \cos^2 x$, $1 + \tan^2 x$, or $1 + \cot^2 x$.
- In implicit differentiation, reuse the *original equation* when it can replace a trig expression with x or y .

Study aid: this is a focused memorization sheet for common Calculus I work, not an official Cengage policy document.